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21 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED
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TH
E MARINE ENVIRONMENT PROTECTION COMMITTEE,
RE
CALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee conferred upon it by international conventions for the prevention and control of marine pollution from ships,
NO
TING that it adopted, by resolution MEPC.328(76), the 2021 revised MARPOL Annex VI, which is expected to enter into force on 1 November 2022 upon its deemed acceptance on 1 May 2022,
NO
TING IN PARTICULAR that the 2021 revised MARPOL Annex VI contains amendments concerning mandatory goal -based technical and operational measures to reduce carbon intensity of international shipping,
NO
TING FURTHER that regulation 23 of MARPOL Annex VI requires that the attained EEXI

concerning mandatory goal -based technical and operational measures to reduce carbon intensity of international shipping,

NO

TING FURTHER that regulation 23 of MARPOL Annex VI requires that the attained EEXI shall be calculated taking into account the guidelines developed by the Organization,

RE

COGNIZING that the aforementioned amendments to MARPOL Annex VI require relevant guidelines for uniform and effective implementation of the regulations and to provide sufficient lead time for industry to prepare,

HA

VING CONSIDERED, at its seventy-sixth session, draft 2021 Guidelines on the method of calculation of the attained Energy Efficiency Existing Ship Index (EEXI),

1 AD

OPTS the 2021 Guidelines on the method of calculation of the attained Energy Efficiency Existing Ship Index (EEXI), as set out in the annex to the present resolution;

2 I

NVITES Administrations to take the annexed Guidelines into account when developing and enacting national laws which give force to and implement requirements set forth in regulation 23 of MARPOL Annex VI;

3 R

EQUESTS the Parties to M ARPOL Annex VI and other Member Governments to

developing and enacting national laws which give force to and implement requirements set forth in regulation 23 of MARPOL Annex VI;

3 REQUESTS the Parties to MARPOL Annex VI and other Member Governments to bring the annexed Guidelines to the attention of masters, seafarers, shipowners, ship operators and any other interested parties;

4 AGREES to keep the Guidelines under review in light of experience gained with their implementation and in light of the review of EEXI regulations to be completed by the Organization by 1 January 2026 as identified in regulation 25.3 of MARPOL Annex VI.

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ANNEX

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1 Definitions

1.1 MARPOL means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocols of 1978 and 1997 relating thereto, as amended.

1.2 For the purpose of these Guidelines, the definitions in MARPOL Annex VI, as amended, apply.

2 Energy Efficiency Existing Ship Index (EEXI)

2.1 EEXI formula

1.2 For the purpose of these Guidelines, the definitions in MARPOL Annex VI, as amended, apply.

2 Energy Efficiency Existing Ship Index (EEXI)

2.1 EEXI formula

The attained Energy Efficiency Existing Ship Index (EEXI) is a measure of ship's energy efficiency (g/t*nm) and calculated by the following formula:

* If part of the Normal Maximum Sea Load is provided by shaft generators, SFCME and CFME may for that part of the power be used instead of SFCAE and CFAE

** In case of P PTI(i) > 0, the average weighted value of (SFCMECFME) and (SFCAECFAE) to be used for calculation of Peff

Note: This formula may not be applicable to a ship having diesel-electric propulsion, turbine propulsion or hybrid propulsion system, except for cruise passenger ships and LNG carriers.

Ships falling into the scope of EEDI requirement can use their attained EEDI calculated in accordance with the 2018 Guidelines on the method of calculation of the attained EEDI for new ships (resolution MEPC.308(73), as amended, the "EEDI Calculation Guidelines" hereafter) as the attained EEXI if the value of the attained EEDI is equal to or less than that of the required EEXI.

new ships (resolution MEPC.308(73), as amended, the "EEDI Calculation Guidelines" hereafter) as the attained EEXI if the value of the attained EEDI is equal to or less than that of the required EEXI.

2.2 Parameters

For calculation of the attained EEXI by the formula in paragraph 2.1, parameters under the EEDI Calculation Guidelines apply, unless expressly provided otherwise. In referring to the aforementioned guidelines, the terminology "EEDI" should be read as "EEXI".

2.2.1 PME(i) ; Power of main engines

In cases where overridable Shaft / Engine Power Limitation is installed in accordance with the 2021 Guidelines on the shaft / engine power limit to comply with the EEXI requirements and use of a power reserve (resolution MEPC.335(76)), PME(i) is 83% of the limited installed power (MCRlim) or 75% of the original installed power (MCR), whichever is lower, for each main engine (i). In cases where the overridable Shaft / Engine Power Limitation and shaft generator(s) are installed, in referring to paragraph 2.2.5.2 (option 1) of the EEDI Calculation Guidelines, "MCRME" should be read as "MCRlim".

engine (i). In cases where the overridable Shaft / Engine Power Limitation and shaft generator(s) are installed, in referring to paragraph 2.2.5.2 (option 1) of the EEDI Calculation Guidelines, "MCRME" should be read as "MCRlim".

For LNG carriers having steam turbine or diesel electric propulsion, PME(i) is 83% of the limited installed power (MCRlim, MPPlim), divided by the electrical efficiency in case of diesel electric propulsion system, for each main engine (i). For LNG carriers, the power from combustion of
( )
mrefwl c i
nef
f
i
MEFMEieffieffAEFAE
nPTI
i
neff
i
iAEeffieffiPTI
n
j
jAEFAEAEiMEiFME
nME
i
iME
n
j
j
f V fCapacity f f f
SFC C P fSFC C P f P fSFC C PSFC C P f

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the excessive natural boil-off gas in the engines or boilers to avoid releasing to the atmosphere

the excessive natural boil-off gas in the engines or boilers to avoid releasing to the atmosphere or unnecessary thermal oxidation should be deducted from PME(i) with the approval of the verifier.

2.2.2 PAE(i) ; Power of auxiliary engines

2.2.2.1 PAE(i) is calculated in accordance with paragraph 2.2.5.6 of the EEDI Calculation Guidelines.

2.2.2.2 For ships where power of auxiliary engines (PAE) value calculated by paragraphs 2.2.5.6.1 to 2.2.5.6.3 of the EEDI Calculation Guidelines is significantly diff erent from the total power used at normal seagoing, e.g. in cases of passenger ships, the PAE value should be estimated by the consumed electric power (excluding propulsion) in conditions when the ship is engaged in a voyage at reference speed (Vref) as given in the electric power table, divided by the average efficiency of the generator(s) weighted by power (see appendix 2 of the EEDI Calculation Guidelines).

2.2.2.3 In cases where the electric power table is not available, the P AE value may be approximated either by:

.1 annual average figure of P AE at sea from onboard monitoring obtained prior

EEDI Calculation Guidelines).

2.2.2.3 In cases where the electric power table is not available, the P AE value may be approximated either by:

.1 annual average figure of P AE at sea from onboard monitoring obtained prior to the EEXI certification;

.2 for cruise passenger ships, approximated value of power of auxiliary engines (PAE,app), as defined below:

, = 0.1193 × + 1814.4 [kW]

.3 for ro-ro passenger ships, approximated value of power of auxiliary engines (PAE,app), as defined below:

, = 0.866 × 0.732 [kW]

2.2.3 Vref ; Ship speed

2.2.3.1 For ships falling into the scope of the EEDI requirement, the ship speed V ref should be obtained from an approved speed-power curve as defined in the 2014 Guidelines on survey and certification of the Energy Efficiency Design Index (EEDI) , as amended (resolution MEPC.254(67), as amended).

2.2.3.2 For ships not falling into the scope of the EEDI requirement, the ship speed Vref should be obtained from an estimated speed-power curve as defined in the 2021 Guidelines on survey and certification of the attained EEXI (resolution MEPC.334(76)).

be obtained from an estimated speed-power curve as defined in the 2021 Guidelines on survey and certification of the attained EEXI (resolution MEPC.334(76)).

2.2.3.3 For ships not falling into the scope of the EEDI requirement but whose sea trial results, which may have been calibrated by the tank test, under the EEDI draught and the sea condition as specified in paragraph 2.2.2 of the EEDI Calculation Guidelines are included in the sea trial report, the ship speed Vref may be obtained from the sea trial report:

= , ×

,

1
3
[knot]

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where,

VS,EEDI, is the sea trial service speed under the EEDI draught; and

PS,EEDI is power of the main engine corresponding to VS,EEDI.

2.2.3.4 For containerships, bulk carriers or tankers not falling into the scope of the EEDI requirement but whose sea trial results, which may have been calibrated by the tank test, under

2.2.3.4 For containerships, bulk carriers or tankers not falling into the scope of the EEDI requirement but whose sea trial results, which may have been calibrated by the tank test, under the design load draught and sea condition as specified in paragraph 2.2.2 of the EEDI Calculation Guidelines are included in the sea trial report, the ship speed Vref may be obtained from the sea trial report:

$$V_{ref} = \frac{1}{3} \times \left(\frac{2}{9} \times \left(\frac{DWT_{S,service}}{k} \times \frac{P_{S,service}}{1} \right)^{\frac{1}{3}} \times V_{S,service} \right) \text{ [knot]}$$

where,

$V_{S,service}$ is the sea trial service speed under the design load draught;

$DWT_{S,service}$ is the deadweight under the design load draught;

$P_{S,service}$ is the power of the main engine corresponding to $V_{S,service}$;

k is the scale coefficient, which should be:

.1 0.95 for containerships with 120,000 DWT or less;

.2 0.93 for containerships with more than 120,000 DWT;

.3 0.97 for bulk carrier with 200,000 DWT or less;

.4 1.00 for bulk carrier with more than 200,000 DWT;

.5 0.97 for tanker with 100,000 DWT or less; and

.6 1.00 for tanker with more than 100,000 DWT.

.3 0.97 for bulk carrier with 200,000 DWT or less;

.4 1.00 for bulk carrier with more than 200,000 DWT;

.5 0.97 for tanker with 100,000 DWT or less; and

.6 1.00 for tanker with more than 100,000 DWT.

2.2.3.5 In cases where the speed-power curve is not available or the sea trial report does not contain the EEDI or design load draught condition, the ship speed V ref can be approximated by Vref,app to be obtained from statistical mean of distribution of ship speed and engine power, as defined below:

, = (,) ×

0.75×

1
3
[knot]

For LNG carriers having diesel electric propulsion system and cruise passenger ship having non-conventional propulsion,

, = (,) ×

1
3
[knot]

where,

Vref,avg is a statistical mean of distribution of ship speed in given ship type and

where,

Vref,avg is a statistical mean of distribution of ship speed in given ship type and ship size, to be calculated as follows:

, = ×

wher
e

A, B and C are the parameters given in the appendix;

mV is a performance margin of a ship, which should be 5% of Vref,avg or one knot, whichever is lower; and

MCRavg is a statistical mean of distribution of MCRs for main engines and MPPavg is a statistical mean of distribution of MPPs for motors in given ship type and ship size, to be calculated as follows:

= ×

wher
e

D, E and F are the parameters given in the appendix;

In cases where the overridable Shaft / Engine Power Limitation is installed, the ship speed Vref approximated by Vref,app should be calculated as follows:

, = (,) ×

0.75×

1
3
[knot]

For LNG carriers having diesel electric propulsion system and cruise passenger ship having non- conventional propulsion, the ship speed V ref approximated by V ref,app should be calculated as follows:

0.75×

1
3
[knot]

For LNG carriers having diesel electric propulsion system and cruise passenger ship having non- conventional propulsion, the ship speed V ref approximated by V ref,app should be calculated as follows:

, = , ×
1
3

2.2.3.6 Notwithstanding the above, in cases where the energy saving device * is installed, the effect of the device may be reflected in the ship speed V ref with the approval of the verifier, based on the following methods in accordance with defined quality and technical standards:

.1 sea trials after installation of the device; and/or

.2 dedicated model tests; and/or

.3 numerical calculations.

* Devices that shift the power curve, which results in the change of PP and Vref, as specified in MEPC.1/Circ.815 on 2013 Guidance on treatment of innovative energy efficiency technologies for calculation and verification of the attained EEDI.

the attained EEDI.
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2.2.4 SFC; Certified specific fuel consumption

In cases where overridable Shaft / Engine Power Limitation is installed, the SFC corresponding to the PME should be interpolated by using SFCs listed in an applicable test report included in an approved NOX Technical File of the main engine as defined in paragraph 1.3.15 of the NOX Technical Code.

Notwithstanding the above, the SFC specified by the manufacturer or confirmed by the verifier may be used.

For those engines which do not have a test report included in the NOX Technical File and which do not have the SFC specified by the manufacturer or confirmed by the verifier, the SFC can be approximated by SFCapp defined as follows:

, = 190 [ ]

, = 215 [ ]

2.2
.5 CF ; Conversion factor between fuel consumption and CO2 emission

be approximated by SFCapp defined as follows:

, = 190 [ ]

, = 215 [ ]

2.2
.5 CF ; Conversion factor between fuel consumption and CO2 emission

For those engines which do not have a test report included in the NOX Technical File and which do not have the SFC specified by the manufacturer, the CF corresponding to SFCapp should be defined as follows:

= 3.114 [2/] f or diesel ships (incl. HFO use in practice)

Ot
herwise, paragraph 2.2.1 of the EEDI Calculation Guidelines applies.

2.2.6 Correction factor for ro-ro cargo and ro-ro passenger ships (fjRoRo)

For ro-ro cargo and ro-ro passenger ships, fjRoRo is calculated as follows:

=
1
1 3 ; if fjRoRo > 1 then fj = 1

wher
e the Froude number, , is defined as:

=
0.5144,

where , is the ship design speed corresponding to 75% of MCRME.:

and the exponents , , and are defined as follows:

Ship type Exponent:

Ro-ro cargo ship 2.00 0.50 0.75 1.00

0.5144,

where , is the ship design speed corresponding to 75% of MCRME.:

and the exponents , , and are defined as follows:

Ship type Exponent:

Ro-ro cargo ship 2.00 0.50 0.75 1.00
Ro-ro passenger ship 2.50 0.75 0.75 1.00

2.2.7 Cubic capacity correction factor for ro-ro cargo ships (vehicle carrier) (fcVEHICLE)

For ro-ro cargo ships (vehicle carrier) having a DWT/GT ratio of less than 0.35, the following cubic capacity correction factor, fcVEHICLE, should apply:

=
0.35 0,8

Where DWT is the capacity and GT is the gross tonnage in accordance with the International Convention of Tonnage Measurement of Ships 1969, annex I, regulation 3.

APPENDIX

Parameters to calculate Vref,avg

Ship type A B C
Bulk carrier 10.6585 DWT of the ship 0.02706
Gas carrier 7.4462 DWT of the ship 0.07604
Tanker 8.1358 DWT of the ship 0.05383
Containership 3.2395
DWT of the ship
where DWT 80,000
80,000
where DWT > 80,000
0.18294
General cargo ship 2.4538 DWT of the ship 0.18832
Refrigerated cargo carrier 1.0600 DWT of the ship 0.31518
Combination carrier 8.1391 DWT of the ship 0.05378
LNG carrier 11.0536 DWT of the ship 0.05030
Ro-ro cargo ship (vehicle carrier) 16.6773 DWT of the ship 0.01802
Ro-ro cargo ship 8.0793 DWT of the ship 0.09123
Ro-ro passenger ship 4.1140 DWT of the ship 0.19863
Cruise passenger ship having
non-conventional propulsion 5.1240 GT of the ship 0.12714

Parameters to calculate MCRavg or MPPavg (= D x EF)

Ship type D E F
Bulk carrier 23.7510 DWT of the ship 0.54087
Gas carrier 21.4704 DWT of the ship 0.59522
Tanker 22.8415 DWT of the ship 0.55826

Parameters to calculate MCRavg or MPPavg (= D x EF)

Ship type D E F
Bulk carrier 23.7510 DWT of the ship 0.54087
Gas carrier 21.4704 DWT of the ship 0.59522
Tanker 22.8415 DWT of the ship 0.55826
Containership 0.5042
DWT of the ship
where DWT 95,000
95,000
where DWT > 95,000
1.03046
General cargo ship 0.8816 DWT of the ship 0.92050
Refrigerated cargo carrier 0.0272 DWT of the ship 1.38634
Combination carrier 22.8536 DWT of the ship 0.55820
LNG carrier 20.7096 DWT of the ship 0.63477
Ro-ro cargo ship (vehicle carrier) 262.7693 DWT of the ship 0.39973
Ro-ro cargo ship 37.7708 DWT of the ship 0.63450
Ro-ro passenger ship 9.1338 DWT of the ship 0.91116
Cruise passenger ship having non -
conventional propulsion 1.3550 GT of the ship 0.88664

Calculation of parameters to calculate Vref,avg and MCRavg

Data sources

1 IHS Fairplay (IHSF) database with the following conditions are used.

Ship type Ship size Delivered period

Calculation of parameters to calculate Vref,avg and MCRavg

Data sources

1 IHS Fairplay (IHSF) database with the following conditions are used.

Ship type Ship size Delivered period
Type of
propulsion
systems
Population
Bulk carrier 10,000 DWT
From 1 January 1999
to 1 January 2009
Conventional 2,433
Gas carrier 2,000 DWT Conventional 292
Tanker 4,000 DWT Conventional 3,345
Containership 10,000 DWT Conventional 2,185
General cargo ship 3,000 DWT Conventional 1,673
Refrigerated cargo carrier 3,000 DWT Conventional 53
Combination carrier 4,000 DWT Conventional 3,351
LNG carrier 10,000 DWT Conventional, Non-conventional 185
Ro-ro cargo ship (vehicle carrier) 10,000 DWT Conventional 301
Ro-ro cargo ship 1,000 DWT From 1 January 1998
to 31 December
2010
Conventional 188
Ro-ro passenger ship 250 DWT Conventional 350
Cruise passenger ship having
non-conventional propulsion 25,000 GT From 1 January 1999 to 1 January 2009 Non-conventional 93

2 Data sets with blank/zero "Service speed", "Capacity" and/or Total kW of M/E" are
removed.

Cruise passenger ship having
non-conventional propulsion 25,000 GT From 1 January 1999 to 1 January 2009 Non-conventional 93

2 Data sets with blank/zero "Service speed", "Capacity" and/or Total kW of M/E" are
removed.

3 Ship type is in accordance with table 1 and table 2 of resolution MEPC.231(65)
on 2013 Guidelines for calculation of reference lines for use with the Energy Efficiency Design
Index (EEDI). However, "Gas carrier" does not include "LNG carrier". Parameters for "LNG
carrier" are given separately.

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